

Mathematical induction

* PMI (principle of mathematical induction)

is a method for proving statements of the form $(\forall n \in \mathbb{N}) P(n)$

PMI

if $\left\{ \begin{array}{l} \text{i) } P(1) \text{ is true and} \\ \text{ii) when } P(k) \text{ is true} \\ \text{for some particular } k \in \mathbb{N}, \\ \text{then } P(k+1) \text{ must also} \\ \text{be true.} \end{array} \right. \Rightarrow \forall n \in \mathbb{N} P(n) \text{ is true.}$

PCI (Principle of complete induction)

PCI

if $\left\{ \begin{array}{l} \text{i) } P(1) \text{ is true and} \\ \text{ii) when } k > 1 \text{ and } P(n) \\ \text{is true for every } n < k, \\ \text{then } P(k) \text{ must also} \\ \text{be true.} \end{array} \right. \Rightarrow \forall n \in \mathbb{N} P(n) \text{ is true.}$

Ex: $\sum_{i=1}^n i = \frac{n(n+1)}{2}$

Proof: (PMI)

Base Case: - $P(1)$ is true.

(ii) Assume $P(k)$ is true for some $k \in \mathbb{N}$
Inductive Hypothesis
 or
Induction assumption

$$\begin{aligned} \sum_{i=1}^{k+1} i &= \sum_{i=1}^k i + (k+1) \\ &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{(k+1)(k+2)}{2} \end{aligned}$$

by PMI $\forall n \in \mathbb{N}$, $P(n)$ is true.

Ex: Show that $L^n > 3^n, \forall n \geq 7$

n	L^n	3^n
1	1	3
2	2	9
3	6	27
4	24	81
5	120	243
6	720	729
7	5040	2187

Base: $n=7$

Induction: $L^k > 3^k$

$$\Rightarrow (k+1) L^k > (k+1) 3^k.$$

$$L^{k+1} > 3^{k+1} \quad \text{since for } k \geq 7$$

$k+1 > 3$

We can start induction anywhere

Two ways in which Induction can fail

- 1) Base case can be false
- 2) Inductive step ~~does not~~ work out.

ex: $\sum_{i=1}^n i = \frac{(n+1.5)^2}{2}$

for $n=1$ R.H.S = 1.125 (failure of base case).

Assume true for arbitrary $n=k$

$$\sum_{i=1}^k i = \frac{(k+1.5)^2}{2}$$

$$\begin{aligned}\sum_{i=1}^{k+1} i &= \frac{(k+1.5)^2}{2} + k+1 \\ &= \frac{k^2 + 2.5k + 2.25 + k+1}{2} \\ &= \frac{k^2 + 3k + 3.25}{2} \\ &= \frac{(k+1.5)^2}{2}\end{aligned}$$

[Importance of Base Case].

Base Case Works.

Inductive step does not work.

$$\sum_{i=1}^n i^2 = \frac{(n+1)^2}{4}$$

Base case: $n=1$ LHS = 1 RHS = 1.

Assume true for $n=k$

$$\sum_{i=1}^k i^2 = \frac{(k+1)^2}{4}$$

$$\begin{aligned} \sum_{i=1}^{k+1} i^2 &= \frac{(k+1)^2}{4} + (k+1) \\ &= \frac{(k+1)(k+5)}{4} \end{aligned}$$

5

4

Ex. 2 (PCI)

Prove that every natural number $n > 1$ is either a prime or a product of primes.

More formally, we are supposed to prove.

$$\forall n \in \mathbb{N} \quad P(n)$$

where

$P(n)$ is

$$(n=1) \vee (n \text{ is prime})$$

$$\vee (n \text{ can be factored into primes})$$

Proof : (i) $P(1)$ is true.

(ii) Suppose $k > 1$ and $P(n)$ is for all $n < k$, we must show $P(k)$ is true.

if k is prime then $P(k)$ is true

Otherwise since $k > 1$, $k = p \cdot q$, $p, q \in \mathbb{N}$ and $< k$

By PCI we conclude that
 $\forall n \in \mathbb{N} \quad P(n)$ is true

(6)



PMI states that

- i) if 1 is red and
- ii) if a natural number must be red whenever its immediate predecessor is red then all natural numbers are red.

PCI states that

- i) if 1 is red and
- ii) if a natural number must be red whenever all its predecessors are red, then all natural numbers must be red.



Well ordering principle:-

"Every non-empty subset of $\mathbb{N}$ (Natural Numbers) contains a smallest element. Essentially the set $\mathbb{N}$ is Well ordered."

Consider proving the following.
Summation to be true for all.
positive integers n .

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

Assume that this statement is false for at least one positive integral value of n .

Let S be the set of all positive integers n for which the statement above is false. Using our assumption S is non-empty. Thus by WOP, S must contain a smallest element. Let this smallest element be k .

We know $K \neq 1$
 $\therefore$ given relation is true for $n=1$

$$\Rightarrow K > 1.$$

$$\Rightarrow \underline{K-1} > 0$$

but this formulae has to be true
for $K-1$

$$\Rightarrow \sum_{l=1}^{K-1} l^0 = \frac{(K-1)[(K-1)+1]}{2}$$

$$\underline{\text{Now}} \quad \sum_{l=1}^K l^0 = \sum_{l=1}^{K-1} l^0 + K$$

$$= \frac{(K-1)K}{2} + K$$

$$= \frac{(K-1)K + 2K}{2}$$

$$= \frac{K(K+1)}{2}$$

Contradiction

